

Kuber Shahi

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EDUCATION

University of California, San Diego

Master of Science in Computer Science, AI Specialization, GPA: 4.0/4.0

Coursework: Stat NLP, Computer Vision I, AI Agents, ML Systems, Software Engineering

San Diego, CA

Sep 2024 – June 2026

Ashoka University

Bachelor of Science in Computer Science, Minor in Physics, GPA: 3.87/4.00

Sonipat, India

Aug 2018 – May 2022

PROFESSIONAL EXPERIENCE

Machine Learning Intern

Aug 2025 – Dec 2025

Melio – AI-driven rapid blood diagnostics biotech startup

- Refactored notebook workflows to a modular Vertex AI custom training pipeline, supporting 3+ time series classifiers with parallelized training and optimized Docker images, reducing experiment turnaround by 20%.
- Standardized reproducibility by centralizing training data and artifacts on GCS and integrating Vertex AI Experiments for automated run tracking, cross-experiment comparison, and model versioning.

Data Scientist

Jun 2022 – May 2024

Vayana Network – India's largest supply chain finance fintech

- Built and optimized 10+ distributed PySpark ETL pipelines on AWS EMR/S3 processing 5M+ invoices annually, cutting ETL latency by 20% through efficient transformations and elimination of redundant operations.
- Designed and deployed 20+ Apache Superset MIS dashboards for real-time reporting on invoice flows and business metrics, serving 100+ internal users across cross-functional teams, boosting operational decision-making.
- Developed a Business Network Analysis platform using NetworkX and Louvain community detection, identifying high-value customer networks and contributing to 15% revenue growth quarterly.
- Spearheaded an Entity Resolution system using NLP vectorization, blocking techniques, and web scraping to deduplicate and enrich 100K+ company records annually, improving data accuracy by 30%.

Data Science Intern

Nov 2021 – Mar 2022

Ageless Partners – Anti-aging and longevity wellness company

- Led the design and prototyping of a wearable-based fitness recommendation system, coordinating a team of 3 interns to develop a personalized engine that improved user engagement and retention by 20%.
- Applied causal inference and statistical modeling using Python and Plotly to identify key drivers of user health outcomes, enhancing de-aging research insights by 15%.

RESEARCH EXPERIENCE

Graduate Student Researcher

Jan 2026 – Present

UCSD BIAG Lab

- Advisor: [Prof. Marc Niethammer](#)
- Researching uncertainty quantification for medical image registration; investigating predictive uncertainty vs. registration error via supervised error regression using TorchIO synthetic deformations. [[GitHub](#)]

Privacy Preserving Neural Networks

Aug 2021 – Dec 2021

Capstone Project, Ashoka University

- Advisors: [Prof. Mahavir Jhawar](#) & Prof. Debayan Gupta
- Researched secure multi-party computation (MPC) techniques to enable privacy-preserving neural network training across distributed systems.
- Implemented [SecureNN](#) protocols in C++ and analyzed their effectiveness in ensuring data privacy and computational efficiency for real-world, data-sensitive applications. [[Presentation](#)] | [[Report](#)] | [[GitHub](#)]

Research Intern

May 2021 – Aug 2021

Mphasis Lab, Ashoka

- Advisor: [Prof. Mahavir Jhawar](#)
- Researched and implemented Privacy-Preserving Machine Learning (PPML) protocols such as SecureML and BLAZE in C++, evaluating each protocol's efficiency and security for sensitive datasets. [[GitHub](#)]

- Co-led the development of a secure, optimized PPML protocol for business-specific needs under Prof. Mahavir Jhawar's guidance, improving the protocol's reliability and applicability for real-world solutions.

Secure Machine Learning

Jan 2021 – May 2021

Independent Study Module, Ashoka University

- Advisor: Prof. Debayan Gupta
- Studied the impact of adversarial attacks such as Data Poisoning and Model Evasion on the performance and reliability of machine learning (ML) models.
- Demonstrated the attacks in the *Subpopulation Data Poisoning Attacks* paper to understand the impact of poisoning attacks on real-world machine learning models. [[Presentation](#)] | [[GitHub](#)]

Applied Cryptography

Jan 2021 – May 2021

Independent Study Module, Ashoka University

- Advisor: Prof. Mahavir Jhawar
- Examined and evaluated the security vulnerabilities in email clients that support the two primary forms of end-to-end email encryption: OpenPGP and S/MIME.
- Illustrated the attacks on various email clients outlined in the *Mailto: Me Your Secrets* paper and suggested countermeasures against them. [[Report](#)] | [[GitHub](#)]

KEY PROJECTS

LLM Agent Planning Under Deception | *GPT-OSS-120B, Llama-4-Scout, Python*

Mar 2026

- Evaluated 4 LLM agent architectures across 2 deceptive text environments; trust-based designs achieved 100% task success while 16 hint/no-hint ablation settings showed planning (not reasoning) as the primary bottleneck. [[GitHub](#)]

Efficient English-Maithili Translation | *Machine Translation, Efficient Fine-Tuning*

Nov 2024 – Dec 2024

- Fine-tuned an English-Hindi (76M) model for low-resource Maithili NMT on 68K parallel sentences; compared LoRA vs. full fine-tuning under compute constraints, achieving a comparable chrF++ score of 35. [[GitHub](#)]

Gamified Kanban Board | *Team Project, Agile Development, GitHub Actions*

Nov 2024 – Dec 2024

- Developed a full-stack gamified task management app with React frontend and Node.js backend, featuring reward-based incentives and interactive task cards; set up CI/CD with GitHub Actions, including automated linting, testing, and deployment. [[GitHub](#)]

Headline Generation | *Text Summarization, LLM Benchmarking, Hugging Face*

Mar 2022 – May 2022

- Engineered a summary-to-headline pipeline by fine-tuning Pegasus (568M) on 98K Indian news articles for abstractive headline generation, with training optimized on distributed HPC using PyTorch and CUDA. [[GitHub](#)]

aCERT: Academic Credential Verification | *Web Apps, MERN Stack, Web3*

Apr 2022 – May 2022

- Engineered a React/Express.js app for tamper-proof academic credential verification on Ethereum; deployed Solidity smart contracts using Truffle on the Sepolia testnet with Redux for state management. [[GitHub](#)]

Synthesizing DFAs using RNNs | *Sequence Learning, Language Representation*

Apr 2021 – May 2021

- Implemented an RNN-based model in Python to automate the synthesis of Deterministic Finite Automata (DFAs), demonstrating the model's ability to learn and represent formal languages. [[GitHub](#)]

Age and Gender Detection | *Image Classification, Deep Learning, Model Optimization*

Nov 2020 – Dec 2020

- Designed and optimized CNN models using Keras and TensorFlow for age and gender classification in facial images, comparing traditional and deep learning approaches to achieve high accuracy. [[GitHub](#)]

TEACHING EXPERIENCE

Teaching Assistant

Sep 2025 – Dec 2025

Political Science Department, UCSD

- **Theories of Technology and National Security** (POLI 145), Spring 2026, Prof. Michael F. Joseph, Class Size: 180. *Tasks:* hold office hours, grade assignments and exams, and assist students with technological concepts (both old and emerging) and their impact on state security.
- **Political Inquiry** (POLI 30D), Fall 2025, Prof. Scott Desposato, Class Size: 210. *Tasks:* conduct weekly discussion sections, hold office hours, grade assignments and exams, and assist students with statistical concepts and STATA.

Undergraduate Teaching Assistant

Aug 2020 – May 2021

CS Department, Ashoka University

- **Discrete Mathematics** (CS 1104), Spring 2021, Prof. Subash Bhalla, Class Size: 100. *Tasks:* holding weekly office hours, setting and grading assignments and test papers, and facilitating online lectures.
- **Probability & Statistics** (CS 1208), Monsoon 2020, Prof. Mahavir Jhawar, Class Size: 80. *Tasks:* holding weekly office hours, grading assignments and test papers, and facilitating online lectures.

STEM Facilitator

Jun 2019 – Jul 2019

Maa Anandmayee Memorial School

- Partnered with MakersBox Delhi to establish a STEM lab at the school by leading equipment selection, procurement, and setup while teaching students hands-on projects in cutting-edge technologies like Robotics, IoT, VR, and Lego.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, MongoDB, JavaScript, R, Bash

ML/AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face, LLMs, NLP, CV, RAG, PEFT (LoRA), CUDA

ML Systems: Vertex AI, W&B, Docker, Kubernetes, Airflow, CI/CD

Data & Cloud: PySpark, AWS, GCP, Hive/Hadoop, Delta Lake

Analysis & Tools: Pandas, NumPy, Jupyter, Plotly, Dash, Apache Superset, Power BI, Git, Linux, Selenium

LEADERSHIP & ACTIVITIES

Resident Assistant, Office of Student Affairs (OSA)	2021 – 2022
Peer Mentor, Office of Learning Support (OLS)	2020 – 2021
Event Organizer, Ashoka University International Student Association (AUISA)	2018 – 2020
Ashoka Volleyball Team, Ashoka University	2018 – 2020

AWARDS & SCHOLARSHIPS

Bronze Medal, CS Awards, Ashoka University	Jun 2022
Silver Medal, CS Awards, Ashoka University	Jun 2021
Dean's List, All 8 Semesters, Ashoka University	All 8 Semesters
Full Scholarship on Tuition and Residence, Ashoka University	2018 – 2022